CSE370 : Database Systems Lab

Database Challenge 01

Activity List

Task 1

Create a database named **'The_Office'.**

CREATE DATABASE The_Office;

Task 2

Use that database and create a table **'Employee'** to record details of the Employees in the Office.

CREATE TABLE Employee (
Emp_ID char(4),
Name varchar(50),
Age int,
Role varchar(30),
Salary int,
Joining_Date date);

Task 3

Insert values from the table below in **'Employee'**

Emp_ID	Name	Age	Role	Salary	Joining_Date
E001	Michael Scott	40	Manager	100000	1999-09-20
E002	Jim Harper	30	Sales Executive	60000	2004-09-30
E003	Pam Beesly	28	Receptionist	25000	2003-09-30
E004	Angela Martin	33	Accountant	65000	2005-09-28
E005	Dwight Shrute	32	Assistant Manager	60000	2003-09-30
E006	Kelly Kapoor	29	Marketing Executive	45000	2003-09-30
E007	Andrew Bernard	30	Sales Executive	50000	2007-05-10
E008	Kevin Malone	28	Accountant	60000	2004-10-30

E009	Toby Flender	35	HR Manager	70000	2004-09-30
E010	Phyllis Vance	40	Sales Executive	61000	1999-09-20
E011	Creed Bratton	50	Sales Executive	80000	1980-06-01

INSERT INTO Employee VALUES
('E001', 'Michael Scott', 40, 'Manager', 100000, '1999-09-20'),
('E002', 'Jim Harper', 30, 'Sales Executive', 60000, '2004-09-30'),
('E003', 'Pam Beesly', 28, 'Receptionist', 25000, '2003-09-30'),
('E004', 'Angela Martin', 33, 'Accountant', 65000, '2005-09-28'),
('E005', 'Dwight Shrute', 32, 'Assistant Manager', 60000, '2003-09-30'),
('E006', 'Kelly Kapoor', 29, 'Marketing Executive', 45000, '2003-09-30'),
('E007', 'Andrew Bernard', 30, 'Sales Executive', 50000, '2007-05-10'),
('E008', 'Kevin Malone', 28, 'Accountant', 60000, '2004-10-30'),
('E009', 'Toby Flender', 35, 'HR Manager', 70000, '2004-09-30'),
('E010', 'Phyllis Vance', 40, 'Sales Executive', 61000, '1999-09-20'),
('E011', 'Creed Bratton', 50, 'Sales Executive', 80000, '1980-06-01');

Task 4

Complete all tasks below:

a. Find the name and role of employees whose name starts with 'a' or ends with 'e'

```
MariaDB [practice]> select name,role from employee where name like 'a%' or '%e'
+----------------+-----------------+
| name           | role            |
+----------------+-----------------+
| Angela Martin  | Accountant      |
| Andrew Bernard | Sales Executive |
+----------------+-----------------+
2 rows in set, 5 warnings (0.001 sec)
```

b. Find the details of Employees who have salary between 40000 and 60000

```
MariaDB [practice]> select *
    -> from employee
    -> where salary>40000 and salary<60000;
+--------+----------------+------+---------------------+--------+--------------+
| emp_id | name           | age  | role                | salary | joining_data |
+--------+----------------+------+---------------------+--------+--------------+
| E006   | Kelly Kapoor   |   29 | Marketing Executive |  45000 | 2003-09-30   |
| E007   | Andrew Bernard |   30 | Sales Executive     |  50000 | 2007-05-10   |
+--------+----------------+------+---------------------+--------+--------------+
2 rows in set (0.000 sec)
```

c. Find the details of employees who have joined before the year 2000.

```
MariaDB [practice]> select * from employee where joining_data<'2001-01-01';
+--------+---------------+------+-----------------+--------+--------------+
| emp_id | name          | age  | role            | salary | joining_data |
+--------+---------------+------+-----------------+--------+--------------+
| E001   | Michael Scott |   40 | Manager         | 100000 | 1999-09-20   |
| E010   | Phyllis Vance |   40 | Sales Executive |  61000 | 1999-09-20   |
| E011   | Creed Bratton |   50 | Sales Executive |  80000 | 1980-06-01   |
+--------+---------------+------+-----------------+--------+--------------+
3 rows in set (0.001 sec)
```

d. There will be a 5% raise in salary for all sales executives, as they have done an excellent job last year. Update the table with the new raised salary. Check if the salary was updated.

```
MariaDB [practice]> update employee set salary = (salary*.05)+salary where role lik
e '%Executive';
Query OK, 5 rows affected (0.006 sec)
Rows matched: 5  Changed: 5  Warnings: 0

MariaDB [practice]> select * from employee;
+--------+----------------+------+---------------------+--------+--------------+
| emp_id | name           | age  | role                | salary | joining_data |
+--------+----------------+------+---------------------+--------+--------------+
| E001   | Michael Scott  |   40 | Manager             | 100000 | 1999-09-20   |
| E002   | Jim Harper     |   30 | Sales Executive     |  63000 | 2004-09-30   |
| E003   | Pam Beesly     |   28 | Receptionist        |  25000 | 2003-09-30   |
| E004   | Angela Martin  |   33 | Accountant          |  65000 | 2005-09-28   |
| E005   | Dwight Shrute  |   32 | Assistant Manager   |  60000 | 2003-09-30   |
| E006   | Kelly Kapoor   |   29 | Marketing Executive |  47250 | 2003-09-30   |
| E007   | Andrew Bernard |   30 | Sales Executive     |  52500 | 2007-05-10   |
| E008   | Kevin Malone   |   28 | Accountant          |  60000 | 2004-10-30   |
| E009   | Toby Flender   |   35 | HR Manager          |  70000 | 2004-09-30   |
| E010   | Phyllis Vance  |   40 | Sales Executive     |  64050 | 1999-09-20   |
| E011   | Creed Bratton  |   50 | Sales Executive     |  84000 | 1980-06-01   |
+--------+----------------+------+---------------------+--------+--------------+
11 rows in set (0.000 sec)
```

e. Michael Scott will get a bonus of 20% on his salary for excellent leadership initiatives last year. Calculate his bonus and use the alias ('Michael_Bonus') for the column header. [Note: You should **not** update his salary. Only show the bonus]

```
MariaDB [practice]> select (.2*salary) as 'Michael_Bonus'
    -> from employee
    -> where name = 'Michael Scott';
+---------------+
| Michael_Bonus |
+---------------+
|       20000.0 |
+---------------+
1 row in set (0.001 sec)

MariaDB [practice]>
```

f. Show the details of all employees according to their salary sorted from higher to lower.

```
MariaDB [practice]> select * from employee order by salary desc;
+--------+----------------+------+---------------------+--------+--------------+
| emp_id | name           | age  | role                | salary | joining_data |
+--------+----------------+------+---------------------+--------+--------------+
| E001   | Michael Scott  |   40 | Manager             | 100000 | 1999-09-20   |
| E011   | Creed Bratton  |   50 | Sales Executive     |  84000 | 1980-06-01   |
| E009   | Toby Flender   |   35 | HR Manager          |  70000 | 2004-09-30   |
| E004   | Angela Martin  |   33 | Accountant          |  65000 | 2005-09-28   |
| E010   | Phyllis Vance  |   40 | Sales Executive     |  64050 | 1999-09-20   |
| E002   | Jim Harper     |   30 | Sales Executive     |  63000 | 2004-09-30   |
| E005   | Dwight Shrute  |   32 | Assistant Manager   |  60000 | 2003-09-30   |
| E008   | Kevin Malone   |   28 | Accountant          |  60000 | 2004-10-30   |
| E007   | Andrew Bernard |   30 | Sales Executive     |  52500 | 2007-05-10   |
| E006   | Kelly Kapoor   |   29 | Marketing Executive |  47250 | 2003-09-30   |
| E003   | Pam Beesly     |   28 | Receptionist        |  25000 | 2003-09-30   |
+--------+----------------+------+---------------------+--------+--------------+
11 rows in set (0.001 sec)
```

g. Show the details of all employees according to their age sorted from lower to higher.

```
MariaDB [practice]> select * from employee order by age asc;
+--------+----------------+------+---------------------+--------+--------------+
| emp_id | name           | age  | role                | salary | joining_data |
+--------+----------------+------+---------------------+--------+--------------+
| E003   | Pam Beesly     |   28 | Receptionist        |  25000 | 2003-09-30   |
| E008   | Kevin Malone   |   28 | Accountant          |  60000 | 2004-10-30   |
| E006   | Kelly Kapoor   |   29 | Marketing Executive |  47250 | 2003-09-30   |
| E002   | Jim Harper     |   30 | Sales Executive     |  63000 | 2004-09-30   |
| E007   | Andrew Bernard |   30 | Sales Executive     |  52500 | 2007-05-10   |
| E005   | Dwight Shrute  |   32 | Assistant Manager   |  60000 | 2003-09-30   |
| E004   | Angela Martin  |   33 | Accountant          |  65000 | 2005-09-28   |
| E009   | Toby Flender   |   35 | HR Manager          |  70000 | 2004-09-30   |
| E010   | Phyllis Vance  |   40 | Sales Executive     |  64050 | 1999-09-20   |
| E001   | Michael Scott  |   40 | Manager             | 100000 | 1999-09-20   |
| E011   | Creed Bratton  |   50 | Sales Executive     |  84000 | 1980-06-01   |
+--------+----------------+------+---------------------+--------+--------------+
11 rows in set (0.001 sec)
```

h. Show details of employees whose age is more than 35 and who joined before 2003.

```
MariaDB [practice]> select *
    -> from employee
    -> where age>35 and joining_data < '2003-01-01';
+--------+---------------+------+-----------------+--------+--------------+
| emp_id | name          | age  | role            | salary | joining_data |
+--------+---------------+------+-----------------+--------+--------------+
| E001   | Michael Scott |   40 | Manager         | 100000 | 1999-09-20   |
| E010   | Phyllis Vance |   40 | Sales Executive |  64050 | 1999-09-20   |
| E011   | Creed Bratton |   50 | Sales Executive |  84000 | 1980-06-01   |
+--------+---------------+------+-----------------+--------+--------------+
3 rows in set (0.000 sec)
```

i. Turns out Creed Bratton has been lying about his age, he is actually 80 years old. So he should retire. Delete him from the table.

```
MariaDB [practice]> delete from employee where name = 'Creed Bratton';
Query OK, 1 row affected (0.003 sec)

MariaDB [practice]>
```

j. Find the details of employees who have the word 'executive' in their role.

```
MariaDB [practice]> select * from employee where role like '%Executive';
+--------+----------------+------+--------------------+--------+--------------+
| emp_id | name           | age  | role               | salary | joining_data |
+--------+----------------+------+--------------------+--------+--------------+
| E002   | Jim Harper     |   30 | Sales Executive    |  63000 | 2004-09-30   |
| E006   | Kelly Kapoor   |   29 | Marketing Executive|  47250 | 2003-09-30   |
| E007   | Andrew Bernard |   30 | Sales Executive    |  52500 | 2007-05-10   |
| E010   | Phyllis Vance  |   40 | Sales Executive    |  64050 | 1999-09-20   |
+--------+----------------+------+--------------------+--------+--------------+
4 rows in set (0.000 sec)
```

k. Change the attribute 'Name' to 'Employee_Name'

```
MariaDB [practice]> alter table employee change column name  employee_name varchar(
50);
Query OK, 0 rows affected (0.004 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

l. Add attribute 'Bonus' to the employee table.

```
MariaDB [practice]> alter table employee add column Bonus int;
Query OK, 0 rows affected (0.008 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

m. Delete attribute 'Bonus' from the table.

```
MariaDB [practice]> alter table employee
    -> drop column Bonus;
Query OK, 0 rows affected (0.008 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

n. List the names of different job roles in the office. There should not be any repetition in your list.

```
MariaDB [practice]> select distinct role
    -> from employee;
+---------------------+
| role                |
+---------------------+
| Manager             |
| Sales Executive     |
| Receptionist        |
| Accountant          |
| Assistant Manager   |
| Marketing Executive |
| HR Manager          |
+---------------------+
7 rows in set (0.001 sec)
```